

From Gardner's Capacity Formula to the Pure Sin-Kernel Limit

A bottom-up derivation with intuitive explanations

Tempotron learning-surface program

June 28, 2026

Abstract

The pure sinusoidal kernel

$$K(s) = \cos(\omega_0 s)$$

collapses the continuous-time tempotron into one two-dimensional Fourier coefficient. The time maximum is the radius of this coefficient, so the classifier is not a half-space but a radial disk gate:

$$\|\mathbf{h}\| \geq \vartheta, \quad \mathbf{h} \in \mathbb{R}^2.$$

Starting from Gardner's spherical-perceptron capacity formula, the replica-symmetric distance form gives

$$\frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2} \int_0^\infty (r - \vartheta)^2 r e^{-r^2/2} dr = \frac{1}{2} \left(\vartheta^2 - \sqrt{2\pi} \vartheta + 2 \right).$$

At balanced firing probability, $\vartheta = \sqrt{2 \ln 2}$, hence

$$\alpha_c^{\text{RS}} = \frac{1}{1 + \ln 2 - \sqrt{\pi \ln 2}} = 4.598069969 \dots$$

This is an *RS existence ceiling*; because the firing set is non-convex, findable capacity can be lower.

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1 One-page result map

Object	Formula
Gardner half-space constraint	$y^\mu \frac{\mathbf{w} \cdot \mathbf{x}^\mu}{\sqrt{N}} \geq \kappa$
Gardner capacity	$\frac{1}{\alpha_c(\kappa)} = \int_{-\infty}^{\kappa} Dz (\kappa - z)^2$
Zero-margin anchor	$\alpha_c(0) = 2$
General RS distance form	$\frac{1}{\alpha_c^{\text{RS}}} = \mathbb{E}_{\sigma, \mathbf{z}} \text{dist}(\mathbf{z}, \mathcal{A}_\sigma)^2$
Pure sin features	$V(t) = A \cos \omega_0 t + B \sin \omega_0 t$
Time maximum	$\max_t V(t) = \sqrt{A^2 + B^2}$
Feasible sets	$\mathcal{A}_+ = \{\ \mathbf{h}\ \geq \vartheta\}, \quad \mathcal{A}_- = \{\ \mathbf{h}\ \leq \vartheta\}$
Sin-kernel RS capacity	$\frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2}(\vartheta^2 - \sqrt{2\pi}\vartheta + 2)$
Balanced threshold	$\vartheta_{\text{bal}} = \sqrt{2 \ln 2}$
Balanced capacity	$\alpha_c^{\text{RS}} = 4.598069969 \dots$
Optimized threshold	$\vartheta_* = \sqrt{\pi/2}, \quad \alpha_c^{\text{RS}}(\vartheta_*) = 4.659792366 \dots$

Intuition. Gardner tells us: capacity is controlled by how far a typical random Gaussian field is from the allowed region. The pure sine kernel changes the allowed region from a half-line to a disk/exterior-of-disk. The rest is geometry plus one Rayleigh integral.

2 The original Gardner capacity formula

2.1 Model

Random patterns and labels:

$$\mathbf{x}^\mu \in \mathbb{R}^N, \quad y^\mu \in \{\pm 1\}, \quad \mu = 1, \dots, P, \quad P = \alpha N.$$

Spherical weights:

$$\|\mathbf{w}\|^2 = N.$$

Margin constraints:

$$y^\mu \frac{\mathbf{w} \cdot \mathbf{x}^\mu}{\sqrt{N}} \geq \kappa.$$

Absorb the labels:

$$\boldsymbol{\xi}^\mu = y^\mu \mathbf{x}^\mu, \quad \frac{\mathbf{w} \cdot \boldsymbol{\xi}^\mu}{\sqrt{N}} \geq \kappa.$$

Gardner volume:

$$\mathcal{V}_N(\alpha, \kappa) = \int_{\|\mathbf{w}\|^2=N} d\mu(\mathbf{w}) \prod_{\mu=1}^{\alpha N} \Theta\left(\frac{\mathbf{w} \cdot \boldsymbol{\xi}^\mu}{\sqrt{N}} - \kappa\right). \quad (1)$$

Quenched entropy:

$$S(\alpha, \kappa) = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \ln \mathcal{V}_N(\alpha, \kappa).$$

Capacity is where the typical volume vanishes:

$$S(\alpha_c, \kappa) = 0.$$

2.2 Gardner's RS formula

The replica-symmetric result is

$$\boxed{\frac{1}{\alpha_c(\kappa)} = I_\kappa := \int_{-\infty}^{\kappa} Dz (\kappa - z)^2} \quad (2)$$

with

$$Dz = \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz.$$

The closed form is

$$\boxed{I_\kappa = (1 + \kappa^2)\Phi(\kappa) + \kappa\phi(\kappa)}, \quad (3)$$

where

$$\Phi(\kappa) = \int_{-\infty}^{\kappa} Dz, \quad \phi(\kappa) = \frac{e^{-\kappa^2/2}}{\sqrt{2\pi}}.$$

At zero margin,

$$I_0 = \int_{-\infty}^0 Dz z^2 = \frac{1}{2}, \quad \boxed{\alpha_c(0) = 2}.$$

Source. The formula (2) is Gardner's spherical perceptron storage calculation [2, 3]. At $\kappa = 0$, the result agrees with Cover's exact linear separability theorem [1].

Intuition. A random pattern creates one Gaussian field value z . If $z \geq \kappa$, it is already safe. If $z < \kappa$, the weight vector must be adjusted enough to push it by distance $\kappa - z$. Capacity is the inverse of the average squared push.

3 Gardner as a distance theorem

The half-space formula generalizes cleanly.

Let a pattern produce a finite-dimensional Gaussian field

$$\mathbf{h}^\mu(\mathbf{w}) \in \mathbb{R}^d.$$

Let the label-dependent acceptable set be

$$\mathcal{A}_\sigma \subset \mathbb{R}^d, \quad \sigma \in \{+, -\}.$$

The constraints are

$$\mathbf{h}^\mu(\mathbf{w}) \in \mathcal{A}_{\sigma^\mu}.$$

The RS $q \rightarrow 1$ decomposition is

$$\mathbf{h}^a = \sqrt{q} \mathbf{z} + \sqrt{1-q} \boldsymbol{\eta}^a, \quad \mathbf{z}, \boldsymbol{\eta}^a \sim \mathcal{N}(0, I_d). \quad (4)$$

Write

$$\Delta = 1 - q \downarrow 0.$$

The probability that a noisy copy lands in $\mathcal{A}$ satisfies the Laplace principle

$$\mathbb{P}_{\boldsymbol{\eta}}[\sqrt{1-\Delta} \mathbf{z} + \sqrt{\Delta} \boldsymbol{\eta} \in \mathcal{A}] \asymp \exp\left(-\frac{\text{dist}(\mathbf{z}, \mathcal{A})^2}{2\Delta}\right). \quad (5)$$

The sphere gives the universal entropy divergence

$$+\frac{1}{2\Delta},$$

while the constraints give

$$-\alpha \frac{\mathbb{E}_{\sigma, \mathbf{z}} \text{dist}(\mathbf{z}, \mathcal{A}_{\sigma})^2}{2\Delta}.$$

At the transition the coefficient of $1/(2\Delta)$ vanishes:

$$1 - \alpha_c^{\text{RS}} \mathbb{E}_{\sigma, \mathbf{z}} \text{dist}(\mathbf{z}, \mathcal{A}_{\sigma})^2 = 0.$$

Thus

$$\boxed{\frac{1}{\alpha_c^{\text{RS}}} = \mathbb{E}_{\sigma, \mathbf{z}} [\text{dist}(\mathbf{z}, \mathcal{A}_{\sigma})^2], \quad \mathbf{z} \sim \mathcal{N}(0, I_d).} \quad (6)$$

Check: half-space. For the perceptron,

$$d = 1, \quad \mathcal{A} = [\kappa, \infty),$$

so

$$\text{dist}(z, \mathcal{A})^2 = (\kappa - z)^2 \mathbf{1}_{z < \kappa}.$$

Then (6) gives Gardner's formula (2).

Source. Equation (6) is the $q \rightarrow 1$ geometric form of Gardner's RS saddle. It is the same “distance-to-feasibility” form used in the reduction notes for soft-max and radial tempotron gates [2, 10, 11].

Intuition. Think of $\mathbf{z}$ as the field produced before the last infinitesimal adjustment of the weights. If $\mathbf{z}$ misses the feasible set, the local Gaussian noise must move it by the shortest distance to the set. The squared shortest distance is the energetic price of one pattern.

4 Pure sinusoidal kernel: exact collapse to two quadratures

Take the pure sinusoidal kernel

$$K(s) = \cos(\omega_0 s). \quad (7)$$

For pattern μ , the membrane trace is

$$V^{\mu}(t; \mathbf{w}) = \sum_{i=1}^N w_i \sum_f \cos\left(\omega_0(t - t_{if}^{\mu})\right). \quad (8)$$

Use

$$\cos(a - b) = \cos a \cos b + \sin a \sin b.$$

Define in-phase and quadrature spike features:

$$c_i^\mu = \sum_f \cos(\omega_0 t_{if}^\mu), \quad s_i^\mu = \sum_f \sin(\omega_0 t_{if}^\mu). \quad (9)$$

Then

$$V^\mu(t; \mathbf{w}) = (\mathbf{w} \cdot \mathbf{c}^\mu) \cos(\omega_0 t) + (\mathbf{w} \cdot \mathbf{s}^\mu) \sin(\omega_0 t) \quad (10)$$

$$=: A^\mu(\mathbf{w}) \cos(\omega_0 t) + B^\mu(\mathbf{w}) \sin(\omega_0 t). \quad (11)$$

Therefore, over a window containing at least one full period,

$$\boxed{\max_t V^\mu(t; \mathbf{w}) = \sqrt{(A^\mu)^2 + (B^\mu)^2}.} \quad (12)$$

The decision rule becomes

$$\sqrt{(\mathbf{w} \cdot \mathbf{c}^\mu)^2 + (\mathbf{w} \cdot \mathbf{s}^\mu)^2} > \Theta. \quad (13)$$

Equivalently,

$$(\mathbf{w} \cdot \mathbf{c}^\mu)^2 + (\mathbf{w} \cdot \mathbf{s}^\mu)^2 > \Theta^2. \quad (14)$$

This is a rank-two quadratic gate:

$$\mathbf{w}^\top M^\mu \mathbf{w} > \Theta^2, \quad M^\mu = \mathbf{c}^\mu (\mathbf{c}^\mu)^\top + \mathbf{s}^\mu (\mathbf{s}^\mu)^\top, \quad \text{rank } M^\mu \leq 2.$$

Source. The single-frequency collapse (11)–(12) is the central degeneracy of the pure-sinusoid reduction, and is also the zero-bandwidth endpoint of the Gabor/narrowband reduction [10, 11].

Intuition. Time has disappeared. Scanning over t only rotates the two-vector (A, B) . The maximum is the length of that vector. More periods do not create more features; they only repeat the same circle.

5 The Gaussian law of the two quadratures

Let the spike phases be

$$\psi_{if}^\mu = \omega_0 t_{if}^\mu \pmod{2\pi}.$$

Assume random phases:

$$\psi_{if}^\mu \sim \text{Unif}[0, 2\pi].$$

Then

$$\mathbb{E}[\cos \psi] = \mathbb{E}[\sin \psi] = 0,$$

$$\mathbb{E}[\cos^2 \psi] = \mathbb{E}[\sin^2 \psi] = \frac{1}{2}, \quad \mathbb{E}[\sin \psi \cos \psi] = 0.$$

If the mean spike count per afferent is $\bar{n}$, then

$$\text{Var}(c_i^\mu) = \text{Var}(s_i^\mu) = \frac{\bar{n}}{2}, \quad \text{Cov}(c_i^\mu, s_i^\mu) = 0.$$

Normalize

$$\tilde{c}_i^\mu = \sqrt{\frac{2}{\bar{n}}} c_i^\mu, \quad \tilde{s}_i^\mu = \sqrt{\frac{2}{\bar{n}}} s_i^\mu.$$

Define normalized fields

$$h_1^\mu(\mathbf{w}) = \frac{\mathbf{w} \cdot \tilde{\mathbf{c}}^\mu}{\sqrt{N}}, \quad h_2^\mu(\mathbf{w}) = \frac{\mathbf{w} \cdot \tilde{\mathbf{s}}^\mu}{\sqrt{N}}. \quad (15)$$

Since $\|\mathbf{w}\|^2 = N$,

$$\text{Var}(h_1^\mu) = \text{Var}(h_2^\mu) = 1, \quad \text{Cov}(h_1^\mu, h_2^\mu) = 0.$$

By the CLT over afferents,

$$\boxed{\mathbf{h}^\mu(\mathbf{w}) = (h_1^\mu, h_2^\mu) \sim \mathcal{N}(0, I_2).} \quad (16)$$

The normalized threshold is

$$\vartheta = \frac{\Theta}{\sqrt{N\bar{n}/2}}.$$

Thus the decision is simply

$$\boxed{\|\mathbf{h}^\mu(\mathbf{w})\| > \vartheta.} \quad (17)$$

Intuition. For random spike phases, the sine and cosine projections are like two independent coin-flip sums. After normalization they become two independent standard Gaussians. The neuron asks whether the resulting point in the plane is outside a circle.

6 The disk geometry

For a firing label,

$$\mathcal{A}_+ = \{\mathbf{h} \in \mathbb{R}^2 : \|\mathbf{h}\| \geq \vartheta\}. \quad (18)$$

For a silent label,

$$\mathcal{A}_- = \{\mathbf{h} \in \mathbb{R}^2 : \|\mathbf{h}\| \leq \vartheta\}. \quad (19)$$

The radius

$$R = \|\mathbf{z}\|, \quad \mathbf{z} \sim \mathcal{N}(0, I_2),$$

has Rayleigh density

$$\boxed{p_R(r) = r e^{-r^2/2}, \quad r \geq 0.} \quad (20)$$

The radial projection distances are

$$\text{dist}(\mathbf{z}, \mathcal{A}_+)^2 = (\vartheta - r)^2 \mathbf{1}_{r < \vartheta}, \quad (21)$$

$$\text{dist}(\mathbf{z}, \mathcal{A}_-)^2 = (r - \vartheta)^2 \mathbf{1}_{r > \vartheta}. \quad (22)$$

Intuition. The silent class is convex: a disk. The firing class is the outside of a disk: non-convex in the induced weight constraints. This is why the formula below should be read as an RS existence estimate/ceiling, not automatically as a guarantee that simple gradient descent will find all solutions.

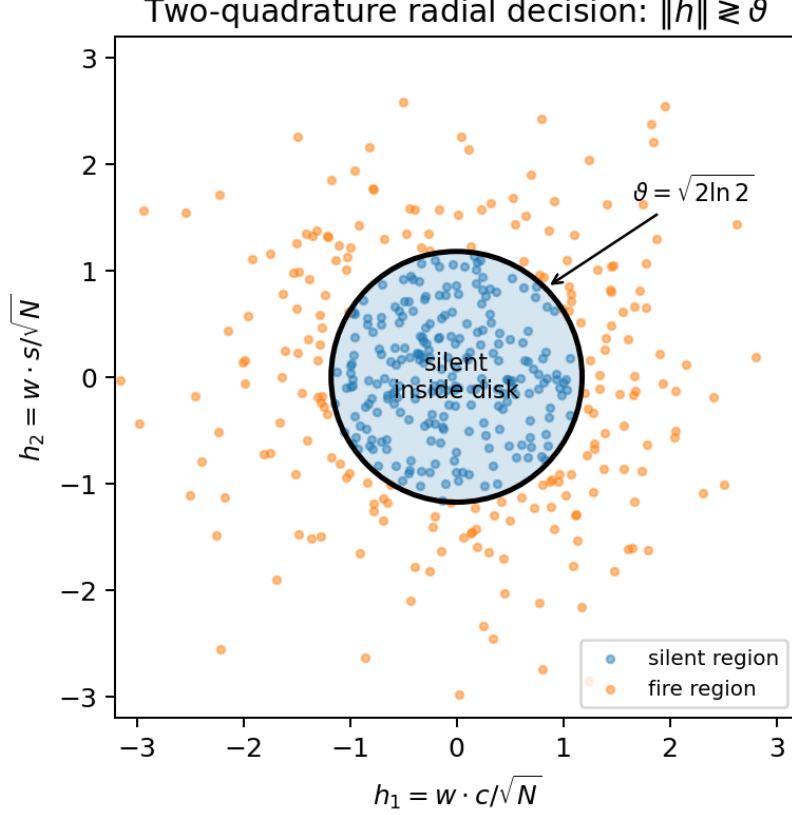


Figure 1: The pure sin-kernel gate is radial. Silent patterns must lie inside the disk, firing patterns outside the disk. The boundary is the balanced threshold $\vartheta = \sqrt{2 \ln 2}$.

7 Apply Gardner's distance theorem

For balanced labels,

$$\mathbb{P}(\sigma = +) = \mathbb{P}(\sigma = -) = \frac{1}{2}.$$

Insert (21)–(22) into the distance theorem (6):

$$\frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2} \int_0^\vartheta (\vartheta - r)^2 r e^{-r^2/2} dr + \frac{1}{2} \int_\vartheta^\infty (r - \vartheta)^2 r e^{-r^2/2} dr. \quad (23)$$

Because the square is the same on both sides,

$$\boxed{\frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2} \int_0^\infty (r - \vartheta)^2 r e^{-r^2/2} dr.} \quad (24)$$

Use Rayleigh moments

$$\mathbb{E}[R] = \sqrt{\frac{\pi}{2}}, \quad \mathbb{E}[R^2] = 2.$$

Then

$$\int_0^\infty (r - \vartheta)^2 r e^{-r^2/2} dr = \mathbb{E}[(R - \vartheta)^2] \quad (25)$$

$$= \mathbb{E}[R^2] - 2\vartheta\mathbb{E}[R] + \vartheta^2 \quad (26)$$

$$= 2 - 2\vartheta\sqrt{\frac{\pi}{2}} + \vartheta^2. \quad (27)$$

Therefore

$$\boxed{\frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2} \left(\vartheta^2 - \sqrt{2\pi} \vartheta + 2 \right)}. \quad (28)$$

Source. The disk-distance integral (24) and closed form (28) are the pure-sinusoid RS capacity calculation in the reduction notes [10]. The derivation is Gardner's distance theorem applied to the radial feasible sets.

Intuition. Every pattern contributes a random point z in the plane. If the point is on the wrong side of the circle, the required correction is exactly the radial gap $|R - \vartheta|$. Capacity is the inverse of the average squared gap, averaged over the two labels.

8 Balanced threshold and the number 4.598

Balanced firing means

$$\mathbb{P}(R > \vartheta_{\text{bal}}) = \frac{1}{2}.$$

For a Rayleigh radius,

$$\mathbb{P}(R > \vartheta) = e^{-\vartheta^2/2}.$$

Therefore

$$e^{-\vartheta_{\text{bal}}^2/2} = \frac{1}{2},$$

so

$$\boxed{\vartheta_{\text{bal}} = \sqrt{2 \ln 2}}. \quad (29)$$

Substitute (29) into (28):

$$\frac{1}{\alpha_c^{\text{RS}}} = \frac{1}{2} \left(2 \ln 2 - \sqrt{2\pi} \sqrt{2 \ln 2} + 2 \right) \quad (30)$$

$$= 1 + \ln 2 - \sqrt{\pi \ln 2}. \quad (31)$$

Hence

$$\boxed{\alpha_c^{\text{RS}}(\vartheta_{\text{bal}}) = \frac{1}{1 + \ln 2 - \sqrt{\pi \ln 2}} = 4.598069969 \dots}. \quad (32)$$

Numerically,

$$\begin{aligned} \vartheta_{\text{bal}} &= 1.1774100225 \dots, \\ 1 + \ln 2 - \sqrt{\pi \ln 2} &= 0.2174825539 \dots, \\ \alpha_c^{\text{RS}} &= 4.5980699691 \dots \end{aligned}$$

9 Threshold optimization

The inverse capacity is

$$C(\vartheta) = \frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2} \left(\vartheta^2 - \sqrt{2\pi}\vartheta + 2 \right).$$

Differentiate:

$$C'(\vartheta) = \vartheta - \sqrt{\frac{\pi}{2}}.$$

The threshold that maximizes capacity is therefore

$$\boxed{\vartheta_* = \sqrt{\frac{\pi}{2}}.} \quad (33)$$

At this value,

$$C(\vartheta_*) = 1 - \frac{\pi}{4}, \quad \boxed{\alpha_c^{\text{RS}}(\vartheta_*) = \frac{1}{1 - \pi/4} = 4.659792366 \dots} \quad (34)$$

Balanced is close to optimal:

$$\frac{\alpha_c^{\text{RS}}(\vartheta_{\text{bal}})}{\alpha_c^{\text{RS}}(\vartheta_*)} = 0.986753 \dots$$

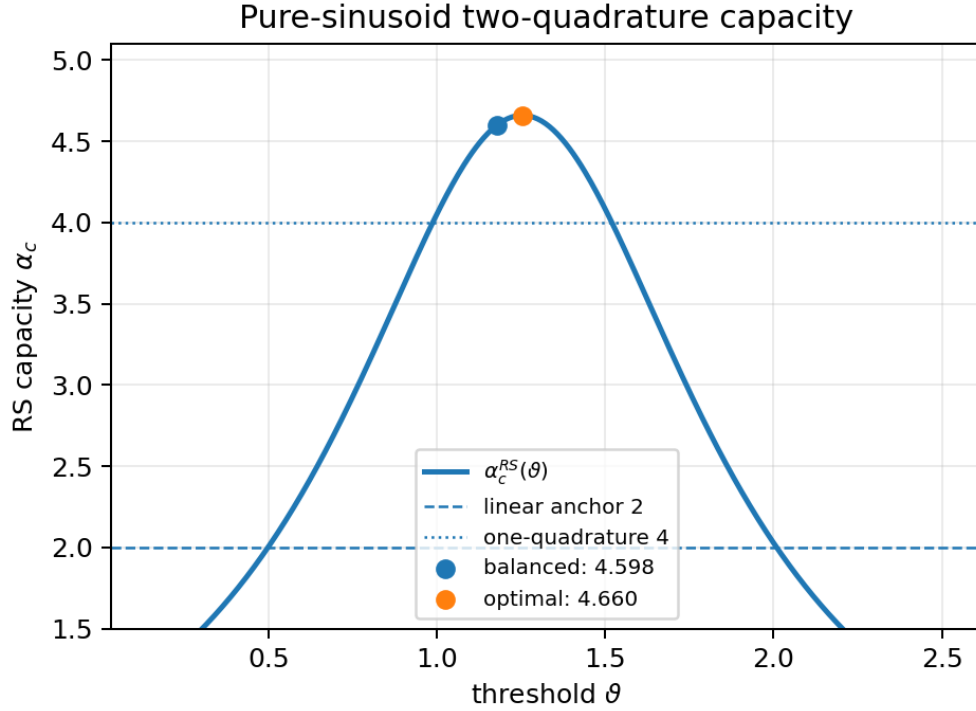


Figure 2: RS capacity as a function of radial threshold. The balanced threshold gives 4.598, close to the optimized ceiling 4.660. The horizontal line at 2 is the linear Cover/Gardner anchor.

Intuition. The balanced threshold is the median radius of a two-dimensional Gaussian. The capacity-optimal threshold is slightly larger because moving the circle outward reduces the average squared correction more efficiently than strict class balance does.

10 Why the pure sin limit is flat in K_{eff}

A finite-band kernel has many Fourier modes. A flat band of width $\Delta\omega$ contains a time-bandwidth count

$$K_{\text{eff}} \sim \Delta\omega T.$$

That count feeds the EVT/cluster lift

$$\Delta\alpha_c \sim \frac{\ln \ln K_{\text{eff}}}{2 \ln 2}$$

for broad or band-limited max-over-time tempotrons [5, 12, 13].

For the pure sinusoid,

$$\Delta\omega = 0.$$

There is only one quadrature pair:

$$\text{span}\{\cos \omega_0 t, \sin \omega_0 t\}.$$

Therefore the capacity is not a growing function of window length once the window contains a period:

$$\alpha_{c,\text{pure sin}}^{\text{RS}} \approx 4.598 \quad \text{is independent of } \omega_0 T \quad (\omega_0 T \gg 2\pi). \quad (35)$$

Intuition. For a broad-band process, more time gives more independent “looks.” For a pure sine wave, more time just repeats the same oscillation. There is only amplitude and phase, i.e. two coordinates.

11 Existence, findability, and the non-convex caveat

The formula

$$\alpha_c^{\text{RS}} = 4.598 \dots$$

is an RS existence calculation.

The firing feasible set is

$$\mathcal{A}_+ = \{\mathbf{h} : \|\mathbf{h}\| \geq \vartheta\},$$

which is non-convex. In weight space, each firing constraint is

$$(\mathbf{w} \cdot \mathbf{c}^\mu)^2 + (\mathbf{w} \cdot \mathbf{s}^\mu)^2 > \Theta^2,$$

the exterior of a rank-two quadratic tube. This can create many disconnected solution regions.

Thus one should separate

$$\alpha_c^{\text{findable}} \leq \alpha_c^{\text{true}} \leq \alpha_c^{\text{RS}}.$$

The project notes report findable values below the RS ceiling in the pure-sinusoid/narrowband regime, with additional restarts improving the crossing but not reaching 4.598 [10, 11].

Intuition. The RS formula asks whether solutions should exist in the large random geometry. A gradient-based learner asks whether a particular search path can find one. Non-convex disk-exterior constraints can make these different.

A Heavy computation A: Gardner half-space integral

Start from

$$I_\kappa = \int_{-\infty}^{\kappa} Dz (\kappa - z)^2.$$

Expand:

$$I_\kappa = \kappa^2 \int_{-\infty}^{\kappa} Dz - 2\kappa \int_{-\infty}^{\kappa} z Dz + \int_{-\infty}^{\kappa} z^2 Dz. \quad (36)$$

The three Gaussian identities are

$$\begin{aligned} \int_{-\infty}^{\kappa} Dz &= \Phi(\kappa), \\ \int_{-\infty}^{\kappa} z Dz &= -\phi(\kappa), \\ \int_{-\infty}^{\kappa} z^2 Dz &= \Phi(\kappa) - \kappa\phi(\kappa). \end{aligned}$$

Therefore

$$I_\kappa = \kappa^2 \Phi(\kappa) + 2\kappa\phi(\kappa) + \Phi(\kappa) - \kappa\phi(\kappa) \quad (37)$$

$$= (1 + \kappa^2)\Phi(\kappa) + \kappa\phi(\kappa). \quad (38)$$

At $\kappa = 0$,

$$I_0 = \Phi(0) = \frac{1}{2}, \quad \alpha_c(0) = \frac{1}{I_0} = 2.$$

B Heavy computation B: Rayleigh moments

The two-dimensional Gaussian radius satisfies

$$R = \sqrt{Z_1^2 + Z_2^2}, \quad Z_1, Z_2 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

The density is

$$p_R(r) = r e^{-r^2/2}, \quad r \geq 0.$$

For $m > -2$,

$$\mathbb{E}[R^m] = \int_0^\infty r^m r e^{-r^2/2} dr. \quad (39)$$

Let

$$u = \frac{r^2}{2}, \quad r dr = du.$$

Then

$$\mathbb{E}[R^m] = \int_0^\infty (2u)^{m/2} e^{-u} du \quad (40)$$

$$= 2^{m/2} \Gamma\left(1 + \frac{m}{2}\right). \quad (41)$$

Thus

$$\begin{aligned}\mathbb{E}[R] &= 2^{1/2}\Gamma(3/2) = \sqrt{\frac{\pi}{2}}, \\ \mathbb{E}[R^2] &= 2\Gamma(2) = 2.\end{aligned}$$

Therefore

$$\mathbb{E}[(R - \vartheta)^2] = \mathbb{E}[R^2] - 2\vartheta\mathbb{E}[R] + \vartheta^2 \quad (42)$$

$$= 2 - 2\vartheta\sqrt{\frac{\pi}{2}} + \vartheta^2 \quad (43)$$

$$= \vartheta^2 - \sqrt{2\pi}\vartheta + 2. \quad (44)$$

Balanced labels multiply by 1/2:

$$\frac{1}{\alpha_c^{\text{RS}}(\vartheta)} = \frac{1}{2}\mathbb{E}[(R - \vartheta)^2] = \frac{1}{2}(\vartheta^2 - \sqrt{2\pi}\vartheta + 2).$$

C Heavy computation C: balanced substitution

Balance gives

$$\mathbb{P}(R > \vartheta) = e^{-\vartheta^2/2} = \frac{1}{2}.$$

Therefore

$$\vartheta^2 = 2 \ln 2, \quad \vartheta = \sqrt{2 \ln 2}.$$

Substitute into the inverse capacity:

$$\frac{1}{\alpha_c^{\text{RS}}} = \frac{1}{2} \left(2 \ln 2 - \sqrt{2\pi} \sqrt{2 \ln 2} + 2 \right) \quad (45)$$

$$= \ln 2 - \sqrt{\pi \ln 2} + 1. \quad (46)$$

Hence

$$\alpha_c^{\text{RS}} = \frac{1}{1 + \ln 2 - \sqrt{\pi \ln 2}}.$$

Numerical values:

$$\begin{aligned}\ln 2 &= 0.6931471805599453, \\ \sqrt{\pi \ln 2} &= 1.475664626635551, \\ 1 + \ln 2 - \sqrt{\pi \ln 2} &= 0.217482553924394, \\ \alpha_c^{\text{RS}} &= 4.59806996908769.\end{aligned}$$

D Heavy computation D: why the $q \rightarrow 1$ saddle gives distance

For fixed $\mathbf{z}$ and small Δ , define

$$Y_\Delta = \sqrt{1 - \Delta} \mathbf{z} + \sqrt{\Delta} \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(0, I_d).$$

The density of Y_Δ at $\mathbf{y}$ is

$$p_\Delta(\mathbf{y} \mid \mathbf{z}) = (2\pi\Delta)^{-d/2} \exp\left[-\frac{\|\mathbf{y} - \sqrt{1 - \Delta}\mathbf{z}\|^2}{2\Delta}\right].$$

The feasibility probability is

$$H_{\mathcal{A}}(\mathbf{z}, \Delta) = \int_{\mathcal{A}} p_\Delta(\mathbf{y} \mid \mathbf{z}) d\mathbf{y}.$$

As $\Delta \downarrow 0$, the integral is dominated by the point of $\mathcal{A}$ closest to $\mathbf{z}$:

$$\inf_{\mathbf{y} \in \mathcal{A}} \|\mathbf{y} - \sqrt{1 - \Delta}\mathbf{z}\|^2 = \text{dist}(\mathbf{z}, \mathcal{A})^2 + O(\Delta).$$

Thus

$$\ln H_{\mathcal{A}}(\mathbf{z}, \Delta) = -\frac{\text{dist}(\mathbf{z}, \mathcal{A})^2}{2\Delta} + O(\ln \Delta).$$

The RS free entropy near capacity has the structure

$$S(\alpha, q) = \frac{1}{2(1 - q)} - \alpha \frac{\mathbb{E} \text{dist}(\mathbf{z}, \mathcal{A}_\sigma)^2}{2(1 - q)} + O(\ln(1 - q)).$$

A non-negative entropy requires the divergent coefficient not be negative. The boundary is

$$1 - \alpha \mathbb{E} \text{dist}(\mathbf{z}, \mathcal{A}_\sigma)^2 = 0.$$

Therefore

$$\frac{1}{\alpha_{\text{RS}}^c} = \mathbb{E} \text{dist}(\mathbf{z}, \mathcal{A}_\sigma)^2.$$

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